GDC

GDC Resolve Encoder

Technical whitepaper and complete configuration guide

H.264 and H.265 codecs (x264 / x265), 10-bit, 4:2:2, HDR10 and own formats for DaVinci Resolve Studio

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Informational document. Export values are starting points; check each platform's current specifications before delivery.

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1. Executive summary

GDC Resolve Encoder is an export plugin for **DaVinci Resolve Studio** that adds software **H.264 (x264)** and **H.265 (x265)** encoders to the codec list, with fine quality control, **10-bit** export, **4:2:2** chroma, **HDR10 metadata** (mastering display, MaxCLL, MaxFALL) and three **own formats** (GDC QuickTime, GDC MP4, GDC Matroska) that write the HDR information straight into the file, not only into the video stream.

The plugin does not replace Resolve's native codecs (ProRes, DNxHR, XDCAM, DCP, EXR). It complements them where you need **controlled H.264/H.265**: deliveries for YouTube, social networks, TVs, streaming platforms, light masters and review copies. Color is not reinterpreted: the plugin receives the finished, graded image from Resolve and encodes it, tagging it correctly (Rec.709, Rec.2020, P3-D65, P3-DCI, PQ, HLG).

At a glance

Aspect	What it offers
Type	Export plugin (IO Encode Plugin) for DaVinci Resolve Studio 21.x
Encoders	x264 and x265 (software); Apple VideoToolbox (Mac); NVIDIA NVENC (Windows with an NVIDIA GPU)
Bit depth	8-bit and 10-bit
Chroma	4:2:0 (8 and 10-bit) and 4:2:2 (10-bit)
Color	Rec.709, Rec.2020, P3-D65, P3-DCI; SDR, PQ (HDR10) and HLG transfer
HDR	Static HDR10: mastering display, MaxCLL, MaxFALL (in the stream and, in the GDC formats, in the container)
Controls	Preset, Profile, Level, Tune, CRF / bitrate / constant QP, keyframe interval, expert x264/x265 parameters
Formats	Resolve's QuickTime, MP4 and MKV + GDC QuickTime, GDC MP4, GDC Matroska
Platforms	macOS on Apple Silicon (arm64), minimum macOS 26 declared by the build, verified on macOS 27. Windows is not part of edition 1.8.0.
Dependencies	Nothing to install: FFmpeg and the required libraries ship inside the package

> **Note.** This document describes what the plugin does, how it works and how to configure it for concrete destinations. Chapter 14 states explicitly what has been verified in DaVinci Resolve and what has not been tested yet.

2. What GDC Resolve Encoder is

2.1 The problem it solves

A colorist or editor eventually faces the same question: which settings should I export with so the file looks good, has a reasonable size and is recognized correctly by the destination platform? H.264 and H.265 have dozens of parameters, and the difference between a good and a poor setting shows up as banding, lost detail, file size and, for HDR, whether the TV or platform recognizes the image as HDR at all.

The plugin brings the widely used x264 and x265 encoders into Resolve, with their parameters exposed in the export panel, plus 10-bit and 4:2:2 export and correct HDR10 metadata.

2.2 What it adds

- **Fine quality control**: CRF (constant quality), target bitrate or constant QP, with preset, level and tune.
- **10-bit** for HDR and for SDR without banding, plus **4:2:2** for masters and deliveries with full vertical chroma.
- **Correct color tags** in the file, read from the project settings: Rec.709, Rec.2020, P3-D65, P3-DCI, PQ, HLG.
- **Static HDR10**: mastering display and MaxCLL/MaxFALL, written into the video stream and, through the GDC formats, into the container.
- **Own formats** (GDC QuickTime, GDC MP4, GDC Matroska) selectable directly in Deliver's Format list.
- **Expert parameters**: a field to pass x264/x265 options directly for special cases.

2.3 Who it is for

- Colorists and editors delivering to **YouTube, Instagram, Facebook, TikTok, Vimeo** who want control over quality.
- Anyone producing **HDR10 or HLG content** who needs the metadata to reach the file.
- Anyone delivering **files for TV playback** (USB, streamer, Apple TV) who wants maximum compatibility.
- Studios that want light **H.264/H.265 4:2:2 10-bit masters** or consistent review copies.

2.4 Requirements and compatibility

Requirement	Details
DaVinci Resolve	**Studio** version (the export plugin SDK works in Studio). Verified on Studio 21.1.
Mac	macOS on **Apple Silicon** (M1/M2/M3/M4...). Intel Macs are not a target of this version.
Windows	Not part of the macOS edition 1.8.0. The free Windows edition, with no activation, is v1.7.1 (maintenance: v1.7.0 with the licence code removed, no new features): the package is yet to be built in CI and has not been verified in DaVinci Resolve on Windows; contributions, including the test, are welcome. NVENC hardware encoding would require an NVIDIA GPU and a current driver.
Linux	Not offered.
Mac installation	Signed and Apple-notarized `.pkg`, installed by double-click into Resolve's IOPlugins folder.
Windows installation	Archive with an install script; copies the plugin into Resolve's IOPlugins folder.
Dependencies	FFmpeg and the required libraries are included in the package; nothing else to install.

Requirement	Details
License	Free and open source: no activation, account or mandatory donation (see chapter 3.6).

3. How it works

3.1 The flow of an export

When you pick a GDC codec in Deliver and start rendering, Resolve processes the timeline (grading, effects, scaling) and hands each finished frame to the plugin. The plugin encodes it and sends the video packets to the container, which writes them to the file together with the audio.

```
Resolve timeline  →  finished YUV frames
      ↓
  GDC plugin: conversion (16 → 10-bit when needed) + color tags
      ↓
  Encoder: x264 / x265 (software)  or  VideoToolbox / NVENC (hardware)
      ↓
  H.264 / H.265 packets
      ↓
  Container: QuickTime / MP4 / MKV (Resolve)  or  GDC QuickTime / GDC MP4 / GDC Matroska
      ↓
  Final file
```

3.2 What the plugin receives from Resolve

Resolve sends the image as **YUV planes**. At 8-bit each sample takes one byte. At 10-bit, Resolve delivers each sample in a **16-bit** container, with the value shifted by 6 bits (that is, on the full 16-bit scale). The plugin converts to 10-bit with **rounding** and clamping to 1023, so it introduces no loss beyond what the codec itself introduces.

> **Note.** You do not need to do anything about this: the conversion is automatic. It is only useful to know if you analyze files with external tools and wonder why the input values are 16-bit.

3.3 The encoders

Encoder	Type	Strengths	Weaknesses
x264 (H.264)	Software	Excellent quality at low bitrate, very fine control, maximum compatibility	Slower than hardware; 10-bit/4:2:2 H.264 is not played by many TVs
x265 (H.265/HEVC)	Software	Smaller files at the same quality, 10-bit and HDR, 4K	Slowest; some older platforms/devices do not play it
Apple VideoToolbox	Hardware (Mac)	Very fast, suited to previews	Lower quality at the same bitrate than x264/x265; 8-bit 4:2:0 only
NVIDIA NVENC	Hardware (Windows)	Very fast on NVIDIA GPUs	Same as above; only with an NVIDIA GPU

3.4 Color tagging

Every file carries, besides the image, **tags** that tell the player which color space it was encoded in (primaries, transfer function, matrix). The plugin reads them from the Resolve project settings (Output Color Space) and writes them into the file. If Resolve sends no value or an unknown one, **Rec.709** is used: the plugin does not guess the color space from the bit depth.

Output Color Space in Resolve	Primaries	Transfer	Matrix
Rec.709 (Scene)	Rec.709 (1)	Rec.709 (1)	Rec.709 (1)
Rec.2100 ST2084 (PQ)	Rec.2020 (9)	SMPTE ST 2084 / PQ (16)	Rec.2020 nc (9)
Rec.2100 HLG	Rec.2020 (9)	ARIB STD-B67 / HLG (18)	Rec.2020 nc (9)
P3-D65	SMPTE EG 432 / P3-D65 (12)	SMPTE ST 428 (17)	Rec.709 (1)
P3-DCI	SMPTE RP 431 / P3-DCI (11)	SMPTE ST 428 (17)	Rec.709 (1)

> **Note.** The table reflects what Resolve 21.1 sends and what the plugin writes (the values in brackets are the standard CICP codes). For HLG the transfer tag is written, but there is no static HDR10-style metadata.

3.5 The containers

There are two families of formats in Deliver's Format list:

- **Resolve's formats** (QuickTime, MP4, MKV): Resolve packages the file and handles all the audio (PCM, AAC, etc.). The GDC codecs appear here as a codec choice. Resolve's container **does not write** HDR10 metadata into the container; it stays in the video stream.
- **The GDC formats** (GDC QuickTime, GDC MP4, GDC Matroska): the plugin packages the file. It writes the color tags, **mastering display** and **MaxCLL/MaxFALL** straight into the container, and the audio is uncompressed **PCM**.

3.6 License and distribution

Edition 1.8.0 for macOS is **free** and **open source**: it asks for no activation code, Machine ID, account, trial period or donation. The plugin code is MIT-licensed; the Mac package includes FFmpeg, x264 and x265 and is distributed under the GPLv3, with the license texts and the source offer inside the package.

4. What it does and does not do

4.1 What it does

Capability	Details
H.264 and H.265 encoding	x264 and x265 at 8 and 10-bit; 4:2:0 (8/10-bit) and 4:2:2 (10-bit)
Quality control	CRF, target bitrate, constant QP; preset, level, tune, keyframe interval
Expert parameters	Free-form string of x264/x265 options (for example aq-mode, psy-rd, vbv-maxrate)
Color tags	Rec.709, Rec.2020, P3-D65, P3-DCI; SDR, PQ, HLG; Video or Full range
Static HDR10	Mastering display (P3-D65 or Rec.2020, peak luminance), MaxCLL, MaxFALL
Own formats	GDC QuickTime, GDC MP4, GDC Matroska, with HDR in the container
Audio	In the GDC formats: 16/24/32-bit PCM, written unchanged (tested with a 1 kHz tone)
Hardware	VideoToolbox (Mac) and NVENC (Windows, NVIDIA GPU) for fast previews
Compatibility	The codecs stay available in Resolve's own QuickTime, MP4 and MKV as well

4.2 What it does not do

Does not do	What to do instead
ProRes, DNxHR/DNxHD, XDCAM, DCP, EXR, DPX	Use Resolve's native codecs
4:4:4; 4:2:2 at 8-bit	4:2:2 exists only at 10-bit; for 4:4:4 use ProRes 4444 or another native codec
Two-pass encoding	Use CRF or target bitrate with VBV (see chapter 11)
HDR10+ and Dolby Vision	The plugin writes static HDR10 only; HLG is only tagged
AAC / AC-3 in the GDC formats	In the GDC formats audio is PCM; for AAC use Resolve's MP4 or QuickTime with a GDC codec
Start timecode and markers in the GDC formats	Not written yet; in Resolve's formats Resolve's own apply
AV1 and VP9	Not included
Color space conversion	Color management is done by Resolve; the plugin only tags
Strict CBR from the panel	Approximate it with vbv-maxrate/vbv-bufsize in advanced parameters
Linux and Intel Mac	Not targets of this version

5. Advantages and disadvantages

5.1 Advantages

- **Better quality/size ratio than hardware encoders**: x264 and x265 produce smaller files at the same visual quality.
- **10-bit and 4:2:2**: less banding in gradients, cleaner chroma on graphics and colored text.
- **Complete HDR10**: tags + mastering display + MaxCLL/MaxFALL, including in the container (GDC formats).
- **Granular control** without leaving Resolve: preset, level, tune, VBV, expert parameters.
- **No extra installs**: FFmpeg and the libraries ship in the package.
- **Compatible** with your existing workflow: the codecs also work in Resolve's native formats.
- **Reproducible**: the same settings give the same result, useful for repeated deliveries.

5.2 Disadvantages and limits

- **Speed**: x265 and slow presets (slower, veryslow) take a long time; hardware is fast but weaker on quality.
- **Audio**: PCM only in the GDC formats; PCM in MP4 has weak support in some players.
- **Playback compatibility**: 10-bit and 4:2:2 H.264 are not played by many TVs and phones; H.265 is not accepted everywhere.
- **No timecode or markers** in the GDC formats.
- **Requires Resolve Studio** (the free Resolve does not load export plugins).
- **Does not cover** deliveries that require ProRes, DNxHR, XDCAM or DCP; those stay with the native codecs.

5.3 When NOT to use the plugin

- When the destination requires a specific broadcast or cinema codec (ProRes, DNxHR, XDCAM, DCP, DPX/EXR).
- When you need the start timecode and markers in the exported file but want the GDC format: use a Resolve format instead.
- When the destination requires AAC and you also want HDR metadata in the container: right now you cannot have both in a single file.
- When you need Dolby Vision or HDR10+.

6. The codec variants

In Deliver, under Codec you pick **GDC Encoder**, then under **Type** one of the variants below (in the plugin's own GDC formats the same variants appear directly as the codec). Hardware variants appear only if the machine has the respective encoder.

Variant (Type)	Bits	Chroma	Profile	Recommended for
GDC H.264 (Software x264)	8	4:2:0	Baseline / Main / High (High default)	Web, social, TVs, maximum compatibility
GDC H.265 (Software x265)	8	4:2:0	Main	SDR 4K, small files
GDC H.264 (Apple VideoToolbox)	8	4:2:0	—	Fast previews on Mac
GDC H.265 (Apple VideoToolbox)	8	4:2:0	—	Fast previews on Mac
GDC H.264 (NVIDIA NVENC)	8	4:2:0	—	Fast previews on Windows with NVIDIA
GDC H.265 (NVIDIA NVENC)	8	4:2:0	—	Fast previews on Windows with NVIDIA
GDC H.264 10-bit (Software x264 High10)	10	4:2:0	High 10	Banding-free 10-bit SDR; limited TV/phone playback
GDC H.265 10-bit (Software x265 Main10)	10	4:2:0	Main 10	HDR10, HLG, 4K, modern deliveries
GDC H.264 4:2:2 10-bit (Software x264 High 4:2:2)	10	4:2:2	High 4:2:2	Light masters, post-production deliveries
GDC H.265 4:2:2 10-bit (Software x265 Main 4:2:2 10)	10	4:2:2	Main 4:2:2 10 (Rext)	HDR 4:2:2 masters, graphics with colored text

> **Warning.** The 10-bit and 4:2:2 variants are intended for post-production and modern platforms. Many TVs, phones and hardware players do not decode 10-bit or 4:2:2 H.264. For TV playback use 8-bit 4:2:0 (SDR) or H.265 Main10 4:2:0 (HDR).

7. Complete settings reference

The plugin's settings ("Plugin Settings") appear under the codec selection. Those marked "software only" do not appear for the hardware variants.

7.1 Preset (software only)

The preset chooses **how much effort** the encoder spends compressing efficiently. A slower preset produces, at the same CRF, a smaller file or better quality, but takes longer. It does not change "what kind" of image you get, only how well it is compressed.

Preset	Speed	Efficiency	When
ultrafast	Maximum	Very poor (large files)	Tests, emergency previews
superfast	Very high	Poor	Quick review
veryfast	High	Medium-poor	Dailies, working copies
faster	Good	Medium	Working copies with acceptable quality
fast	Good	Medium-good	Quick deliveries
medium (default)	Balanced	Good	General use
slow	Slower	Very good	**Recommended for final delivery**
slower	Slow	Excellent	Final delivery, when you have time
veryslow	Very slow	Maximum	Small gain over "slower"; rarely justified

> **Tip.** Rule of thumb: pick **slow** for final delivery and **veryfast** for review. The difference between slower and veryslow is usually small compared to the extra time.

7.2 Profile (software only, 8-bit H.264)

Profile	What it means	When
baseline	Restricted toolset, no B-frames; lower quality at equal bitrate	Only very old devices
main	Wide compatibility, fewer tools than High	Older players that do not accept High
high (default)	The full 8-bit toolset; best efficiency	Almost any modern destination

For H.265, at 10-bit and at 4:2:2 the profile is chosen automatically by the encoder (Main, Main 10, Main 4:2:2 10, High 10, High 4:2:2).

7.3 Level (software only)

The level limits resolution, frame rate and maximum bitrate so the file can be played by certain devices. **Auto** (default) lets the encoder choose according to resolution; it suits most cases. Set it manually only when a specific device requires a maximum level.

Level	H.264: suited for	H.265: suited for
3.0 / 3.1	SD and 720p	720p
4.0 / 4.1	1080p up to 30 fps	1080p up to 30 fps (4.0), 1080p60 (4.1)
4.2	1080p60	1080p60 and above
5.0	Large resolutions, low frame rates	**4K up to 30 fps**
5.1	**4K up to 30 fps**	**4K up to 60 fps**
5.2	**4K up to 60 fps**	4K up to 120 fps

> **Note.** A level too low for the chosen resolution makes the encoder refuse the combination or limit the bitrate. A level that is too high does no harm, but may make the file undecodable on devices that stop at a lower level.

7.4 Tune (software only)

Tune optimizes the encoder for a certain kind of content. "none" (default) is good in most cases.

Tune	What it does	When	Available
none	No special optimization	General use	H.264, H.265
film	Preserves fine detail on filmed content, high bitrate	Films, finely textured content	H.264 only
animation	Optimized for flat areas, clean edges	Animation, graphics, drawings	H.264, H.265
grain	Preserves grain structure, keeps it uniform	Film-grain material	H.264, H.265
stillimage	Optimized for nearly static images	Presentations, slideshows	H.264 only
psnr / ssim	Optimizes objective metrics, not appearance	Technical comparisons, not delivery	H.264, H.265
fastdecode	Simplifies the stream so it decodes easily	Weak devices; lowers compression efficiency	H.264, H.265
zerolatency	No encoding delay (no B-frames/ lookahead)	Live streaming; not for delivery files	H.264, H.265

> **Warning.** fastdecode and zerolatency reduce compression efficiency: the same quality needs a larger file. Do not use them for ordinary deliveries.

7.5 Rate Control: how you choose quality

You have three modes. The choice determines whether you control **quality** or **size**.

Mode	What you control	Advantage	Disadvantage	When
Constant Quality (CRF) (default)	Quality: number 0–51, lower = better	Constant quality across the film; most efficient	Final size not known beforehand	Almost always
Target Bitrate	Size: kbps (500–100000)	Predictable size	Variable quality: heavy scenes may come out weak	When the platform requires a bitrate; hardware

Mode	What you control	Advantage	Disadvantage	When
Constant QP	Each frame's quantization	Predictable behavior, no adaptation	Inefficient; large files	Tests, analysis, special cases

> **Tip.** The rule: if you have no bitrate requirement, use **CRF**. If the platform imposes a ceiling, use CRF **plus** a VBV cap (see chapter 11).

> **Note.** The hardware variants (VideoToolbox, NVENC) have no native CRF: if you pick CRF, the plugin uses a default bitrate of about 12 Mbps. For hardware pick **Target Bitrate**.

7.6 Quality (CRF): recommended values

The values below are **starting points** for ordinary filmed content; very detailed or grainy content needs a lower CRF. x265 uses a slightly different scale: for the same visual quality, x265's CRF is usually **3–5 units higher** than x264's (rule of thumb).

Purpose	x264 (H.264)	x265 (H.265)	Notes
Master / light archive	14–16	16–18	Large files; 10-bit recommended
High-quality delivery	17–19	19–22	Demanding client, festivals
Standard delivery (YouTube, Vimeo)	18–21	21–24	Platform recompresses; keep headroom
Social (Instagram, TikTok, Facebook)	20–23	23–26	Platform recompresses heavily
Review / proxy	24–28	27–30	Small size, sufficient quality
Very small file	26–30	28–32	Visible loss on detail

7.7 Bit Rate (Target Bitrate)

A slider from **500 to 100000 kbps**, in steps of 100. The values below are indicative for SDR H.264; for H.265 you can use about **60–70%** of these values at comparable quality (rule of thumb).

Resolution / frame rate	H.264 SDR	H.265 / HDR (indicative)
720p 24–30	5–7.5 Mbps	4–6 Mbps
1080p 24–30	8–12 Mbps	6–9 Mbps
1080p 48–60	12–18 Mbps	9–13 Mbps
1440p 24–30	16 Mbps	12 Mbps
4K 24–30	35–45 Mbps	25–35 Mbps (HDR: 44–56 Mbps recommended by some platforms)
4K 48–60	53–68 Mbps	40–50 Mbps (HDR: 66–85 Mbps recommended by some platforms)

7.8 Keyframe Interval (sec)

The distance between two keyframes (complete images), between **1 and 10 seconds**, default **2 seconds**. The plugin converts it into frames using the project's real frame rate. A short interval means fast scrubbing and good error recovery, but a slightly larger file; a long one means a slightly smaller file but slower scrubbing.

Purpose	Value
Streaming / web (default)	2 seconds
Social, platforms that recompress	1–2 seconds
Post-production deliveries, later editing	1 second
Small file for a viewing archive	4–5 seconds

The number of B-frames is fixed at **2**; it cannot be changed from the panel.

7.9 Advanced Params (x264/x265)

A text field where you write raw encoder options in the form `key=value:key=value` (for example `aq-mode=3:psy-rd=1.0,0.15`). They are passed directly to x264 or x265, depending on the chosen variant. Details and examples in chapter 11.

> **Warning.** An invalid string makes the encoder refuse to start the export. If the export does not start after you fill in the field, clear it and try again. Options you write here take priority over those in the panel (including HDR10).

7.10 HDR10 Metadata (software only)

The group of settings that writes static HDR10 metadata. It is **off** by default. It applies only when the export is **PQ** (Output Color Space: Rec.2100 ST2084).

Setting	Values	What it means
HDR10 Metadata	Off / On (PQ exports only)	Off: nothing is written. On: the data below is written, only if the export is PQ.
Mastering Primaries	P3-D65 / Rec.2020	The gamut of the **monitor you graded on** (not the image's color space). P3-D65 is the usual case.
Mastering Peak	100–10000 nits (default 1000)	The maximum luminance of the mastering monitor.
MaxCLL	0–10000 nits (0 = not signalled)	The brightest pixel in the whole film.
MaxFALL	0–4000 nits (0 = not signalled)	The highest frame-average luminance of any frame in the whole film.

The mastering display's minimum luminance is fixed at 0.005 nits. More about HDR in chapter 8.

7.11 Resolve settings that affect the export

Resolve setting	Recommendation	Why
Output Color Space (Color Management)	Rec.709 for SDR; Rec.2100 ST2084 for HDR10; Rec.2100 HLG for HLG	The color tags written into the file come from here
Data Levels (Advanced Settings)	**Video** for almost any delivery	Full range makes some players show the image washed out or too contrasty
Color Space Tag / Gamma Tag	Same as project	Lets the tagging follow the project
Retain sub-black and super-white data	Unchecked for delivery	Only for intermediate streams
Resolution	**Even** width and height	Odd dimensions are refused by the plugin
Frame rate	The timeline's	Changing the rate at export causes stutter
Export Audio	Checked, or unchecked if you work separately	The GDC formats write PCM

7.12 Fixed behaviors

- B-frames: 2. The encoder automatically uses the number of threads suited to the processor (up to 32).
- Encoding does not hold the whole file in memory: frames are encoded one at a time, so memory use stays stable even on long exports.
- NVENC hardware variants retry up to 3 times to start, because the GPU session may be temporarily busy.

8. Color and HDR in detail

8.1 SDR: Rec.709, gamma 2.4

The standard delivery for web and SDR television: Rec.709 primaries, Rec.709/BT.1886 transfer (gamma 2.4 on a calibrated monitor), **Video** level range (16–235 at 8-bit). In Resolve, Output Color Space: Rec.709. The plugin writes tags 1/1/1 and adds no HDR metadata.

8.2 HDR10 (PQ)

HDR10 uses the **PQ (SMPTE ST 2084)** transfer function and **Rec.2020** primaries, at **10-bit**, with static metadata (mastering display, MaxCLL, MaxFALL). It is the HDR format accepted by most TVs and platforms.

- Output Color Space: **Rec.2100 ST2084**.
- Codec: an H.265 **10-bit** variant (Main10) or 4:2:2 10-bit.
- HDR10 Metadata: **On**, with the real mastering monitor.
- Format: **GDC MP4**, **GDC QuickTime** or **GDC Matroska**, so the metadata also reaches the container.

8.3 HLG

HLG (Hybrid Log-Gamma) is used mostly in television because it looks reasonable on SDR screens too. It is exported with Output Color Space **Rec.2100 HLG**; the plugin writes the HLG transfer tag. HLG **does not use** static HDR10 metadata, so set **HDR10 Metadata: Off**.

8.4 P3-D65 and P3-DCI

If you work in P3-D65 (for example for Apple deliveries or P3 screens) or in P3-DCI (digital cinema), the matching Output Color Space makes the plugin write P3 primaries into the stream and the container. Check that the destination platform interprets P3 correctly; many web platforms expect Rec.709 or Rec.2020.

8.5 Mastering display vs the image's color space

These are two different things, often confused:

	The image's color space	Mastering display
What it describes	How the pixels are coded (e.g. Rec.2020 + PQ)	The monitor you graded on
Where it comes from	From Output Color Space in Resolve	From the plugin panel (Mastering Primaries, Mastering Peak)
Who uses it	The player, when decoding and displaying	The TV, for tone mapping
Does it change the pixels?	Yes (it defines their interpretation)	No; it only informs the TV

A common example: the image is in a Rec.2020/PQ container, while the mastering display is **P3-D65** with a **1000-nit** peak, because you graded on a P3 monitor that reaches 1000 nits. Choose **Rec.2020** under Mastering Primaries only if your monitor covers almost the whole Rec.2020 gamut (rare).

8.6 MaxCLL and MaxFALL

- **MaxCLL** (Maximum Content Light Level): the luminance of the brightest pixel in the whole film, in nits.
- **MaxFALL** (Maximum Frame-Average Light Level): the highest average luminance of any frame in the whole film, in nits.

The correct values must be **measured** on the final content. If you have not measured them, leave **0** (not signalled), which is more correct than an invented value: a MaxCLL that is too low or too high can make the TV's tone mapping behave badly.

> **Warning.** The values in this document's examples (1000 and 400) are illustrative only. Do not copy them into real deliveries without measuring.

8.7 Example: grading on a laptop screen with an HDR profile

If you grade on the internal screen of an Apple laptop with the HDR profile active, the screen works in P3 with PQ. Prudent settings: **Mastering Primaries: P3-D65**, **Mastering Peak: 1000**, MaxCLL/MaxFALL measured or 0. A laptop screen is not an HDR reference monitor: for deliveries with strict requirements, also check on an HDR TV. An external SDR monitor calibrated to Rec.709 cannot evaluate HDR; the SDR version is checked separately.

8.8 What HDR metadata is not supported

- **HDR10+** (dynamic metadata) and **Dolby Vision**: not written by the plugin.
- **HLG** is tagged, but has no additional static metadata.

9. Containers and audio

9.1 Resolve's formats vs the GDC formats

	Resolve's formats (QuickTime / MP4 / MKV)	The GDC formats (GDC QuickTime / MP4 / Matroska)
Who packages	Resolve	The plugin (through FFmpeg)
Audio	Any codec Resolve offers (PCM, AAC, etc.)	16/24/32-bit PCM, written unchanged
HDR10 metadata in the container	No	Yes: mastering display and MaxCLL/ MaxFALL
Color tags in the container	Yes, written by Resolve	Yes, written by the plugin
Start timecode / markers	Yes	No (yet)
How it appears in Deliver	Format = QuickTime/MP4/MKV, then Codec = GDC Encoder, then Type	Format = GDC ..., then Type (directly)

The GDC formats appear in the **Format** list after Resolve's native formats, alongside other plugin formats; their position in the list cannot be changed.

9.2 Audio

In the GDC formats, Resolve sends uncompressed audio (PCM) and the plugin writes it unchanged. That is the professional choice for masters and post-production deliveries. For destinations that require AAC, use Resolve's MP4 or QuickTime format with a GDC video codec.

Container	PCM audio	Note
GDC QuickTime (.mov)	Good	Standard in post-production
GDC Matroska (.mkv)	Very good	Wide support for PCM
GDC MP4 (.mp4)	Limited	PCM in MP4 (ipcm) is played poorly by some players (e.g. QuickTime/Safari)

> **Note.** If you do not need audio in the file, uncheck "Export Audio" in Deliver; the container will then receive no audio track.

9.3 Which combination to choose

Purpose	Format	Why
HDR10 with metadata in the file + PCM audio	GDC QuickTime or GDC Matroska	Metadata in the container; PCM well supported
HDR10 for players that require MP4	GDC MP4	Metadata in the container; check audio playback
SDR with AAC for web/social	Resolve's MP4 + GDC codec	AAC is compatible everywhere
Delivery that requires timecode/markers	A Resolve format + GDC codec	Resolve writes them

10. Export recipes

Each recipe lists the settings in the plugin panel and in Resolve. The values are **starting points**: test on a short 10–20 second clip, check the result on the target device, then export the film. Platforms change their recommendations periodically; check their current specifications before an important delivery.

10.1 How to read the recipes

- **Format** = Deliver's Format list. "MP4 (Resolve)" means the native format; "GDC MP4" means the plugin's own format.
- **Codec (Type)** = the variant chosen in the plugin panel.
- **In Resolve** = the Deliver and Color Management settings that affect the file.
- If a row is missing from a recipe, leave the default value.

> **Tip.** For any destination, first export a fragment from the hardest part of the film (fast motion, gradients, darkness). If it looks good there, the rest will look good.

10.2 YouTube

YouTube recompresses everything you upload. The goal is to upload a **high-quality** file so the recompression starts from a clean source. The bitrate values below are the ones YouTube publicly recommended at the time of writing, as an order of magnitude.

YouTube SDR, 1080p (24–30 fps)

Setting	Value
Format	MP4 (Resolve) or QuickTime (Resolve), with AAC 320 kbps or PCM audio
Codec (Type)	GDC H.264 (Software x264)
Preset	slow
Profile / Level	High / Auto (4.2 if the frame rate is 48–60)
Tune	none
Rate Control	Constant Quality, **CRF 18–20** (or Target Bitrate 12 Mbps)
Keyframe Interval	1–2 seconds
HDR10 Metadata	Off
In Resolve	Output Color Space Rec.709; Data Levels Video; 1920×1080; the timeline's frame rate
Keep in mind	YouTube's recommendation for 1080p SDR is on the order of 8 Mbps at 24–30 fps and 12 Mbps at 48–60 fps; a CRF of 18–20 exceeds these values, which is good for recompression.

YouTube SDR, 4K (24–30 fps)

Setting	Value
Format	MP4 (Resolve) with AAC, or GDC QuickTime with PCM
Codec (Type)	GDC H.264 (Software x264) or GDC H.265 (Software x265) for smaller files

Setting	Value
Preset	slow
Profile / Level	H.264: High / **5.1** (5.2 at 60 fps). H.265: Main / **5.0** (5.1 at 60 fps)
Tune	none
Rate Control	H.264: **CRF 18–20** (or 35–45 Mbps). H.265: **CRF 21–23** (or 25–35 Mbps)
Keyframe Interval	1–2 seconds
HDR10 Metadata	Off
In Resolve	Output Color Space Rec.709; Data Levels Video; 3840×2160
Keep in mind	H.265 at 4K reduces the file by about a third compared to H.264 at comparable quality, but the export takes longer.

YouTube HDR10 (PQ), 4K

Setting	Value
Format	**GDC MP4**, **GDC QuickTime** or **GDC Matroska** (HDR10 metadata reaches the container)
Codec (Type)	GDC H.265 10-bit (Software x265 Main10)
Preset	slow
Profile / Level	Main 10 (automatic) / **5.0** (4K 30 fps) or **5.1** (4K 60 fps)
Tune	none
Rate Control	**CRF 16–18** (or Target Bitrate 44–56 Mbps at 4K 24–30 fps, 66–85 Mbps at 48–60 fps)
Keyframe Interval	1–2 seconds
HDR10 Metadata	**On**; Mastering Primaries P3-D65; Mastering Peak = your monitor's luminance (1000 typical); MaxCLL/MaxFALL measured, or 0
In Resolve	Output Color Space **Rec.2100 ST2084**; Data Levels Video; 3840×2160
Audio	24-bit 48 kHz PCM in the GDC formats (check on upload that the platform accepts it)
Keep in mind	YouTube identifies HDR from the file. The GDC formats also put the metadata in the container, which is the safest option. After uploading, check that the platform marks the video as HDR.

YouTube HLG, 4K

Setting	Value
Format	GDC MP4 or GDC QuickTime
Codec (Type)	GDC H.265 10-bit (Software x265 Main10)
Preset / Level	slow / 5.0 (4K 30 fps) or 5.1 (4K 60 fps)
Rate Control	**CRF 16–18**
Keyframe Interval	1–2 seconds
HDR10 Metadata	**Off** (HLG does not use static metadata)
In Resolve	Output Color Space **Rec.2100 HLG**; Data Levels Video
Keep in mind	Check on both an SDR and an HDR screen: HLG looks acceptable on both.

10.3 Facebook, Instagram, TikTok and other networks

Social networks **recompress heavily**. There is no point in uploading a huge file: a good-quality file, in the right frame format and with a moderate bitrate, behaves best. Use **8-bit SDR H.264**, the safest choice. Size, duration and resolution limits change often; check them before uploading.

Frame format	Resolution	Where it is used
Vertical 9:16	1080×1920	Instagram Reels and Stories, TikTok, Facebook Reels, YouTube Shorts
Portrait 4:5	1080×1350	Instagram and Facebook Feed (takes more screen)
Square 1:1	1080×1080	Feed, special cases
Landscape 16:9	1920×1080	Facebook, YouTube, LinkedIn, X

Instagram Reels / Stories (vertical)

Setting	Value
Format	MP4 (Resolve), AAC audio 128–256 kbps, 48 kHz
Codec (Type)	GDC H.264 (Software x264)
Preset	slow (or medium for speed)
Profile / Level	High / 4.1 (Auto is enough)
Tune	none
Rate Control	**CRF 20**; optional cap in Advanced Params: `vbv-maxrate=12000:vbv-bufsize=24000`
Keyframe Interval	2 seconds
HDR10 Metadata	Off
In Resolve	Output Color Space Rec.709; Data Levels Video; 1080×1920; 30 fps (or native rate)
Keep in mind	Keep text and important elements in the central area; the app's interface covers the top and bottom edges.

Instagram Feed / Facebook Feed (4:5 or 1:1)

Setting	Value
Format	MP4 (Resolve), AAC audio 128–192 kbps
Codec (Type)	GDC H.264 (Software x264)
Preset / Profile / Level	slow / High / Auto
Rate Control	**CRF 20–22** (or Target Bitrate 8–10 Mbps at 1080p)
Keyframe Interval	2 seconds
In Resolve	Rec.709; Data Levels Video; 1080×1350 (4:5) or 1080×1080 (1:1)

Facebook (landscape video)

Setting	Value
Format	MP4 (Resolve), AAC stereo audio 128–256 kbps
Codec (Type)	GDC H.264 (Software x264)
Preset / Profile / Level	slow / High / Auto
Rate Control	**CRF 20–22** (or Target Bitrate 8–12 Mbps at 1080p)
Keyframe Interval	2 seconds
In Resolve	Rec.709; Data Levels Video; 1920×1080; frame rates up to 30 fps recommended

TikTok

Setting	Value
Format	MP4 (Resolve), AAC audio 128–192 kbps
Codec (Type)	GDC H.264 (Software x264)
Preset / Profile / Level	slow / High / 4.1
Rate Control	**CRF 20**; alternatively Target Bitrate 8–12 Mbps
Keyframe Interval	2 seconds
In Resolve	Rec.709; Data Levels Video; 1080×1920; 30 fps
Keep in mind	HDR support on upload differs from one app version to another: if you want HDR, test on a short clip before delivering.

Vimeo

Setting	Value
Format	MP4 or QuickTime (Resolve)
Codec (Type)	GDC H.264 (Software x264)
Preset / Profile / Level	slow / High / Auto
Rate Control	**CRF 17–19** (or Target Bitrate 10–20 Mbps at 1080p)
Keyframe Interval	2 seconds
In Resolve	Rec.709; Data Levels Video; the project's resolution

Quick sharing (WhatsApp, Telegram, e-mail)

Setting	Value
Format	MP4 (Resolve), AAC audio 128 kbps
Codec (Type)	GDC H.264 (Software x264)
Preset	fast
Rate Control	**CRF 24–26**
Keyframe Interval	2–4 seconds
In Resolve	Rec.709; Data Levels Video; 1280×720 or 1920×1080

Setting	Value
Keep in mind	Messaging apps recompress and limit size; a small file arrives faster and loses less on recompression.

10.4 Television: playback from USB, streamer or Apple TV

For TV playback, what matters is the **device's hardware decoder**, not the encoder's quality. TVs play H.264 8-bit 4:2:0 almost universally and, on recent models, H.265 Main/Main10 4:2:0. They generally **do not** play 10-bit or 4:2:2 H.264. The safe rule: for TV, stay with 4:2:0.

SDR TV, 1080p (USB / DLNA)

Setting	Value
Format	MP4 (Resolve) with AAC or AC-3 (compatible with most TVs)
Codec (Type)	GDC H.264 (Software x264), 8-bit
Preset / Profile / Level	slow / High / **4.1** (4.2 at 50/60 fps)
Tune	none
Rate Control	**CRF 17–19** (or Target Bitrate 15–25 Mbps)
Keyframe Interval	2 seconds
In Resolve	Rec.709; Data Levels Video; 1920×1080
Keep in mind	Do not use 10-bit or 4:2:2 for TV. If the TV does not play the audio, switch to stereo AAC.

SDR TV, 4K

Setting	Value
Format	MP4 (Resolve) with AAC
Codec (Type)	GDC H.265 (Software x265), 8-bit (or GDC H.264 if the TV does not play H.265)
Preset / Level	slow / **5.0** (30 fps) or **5.1** (60 fps)
Rate Control	**CRF 20–22** (or Target Bitrate 30–50 Mbps)
Keyframe Interval	2 seconds
In Resolve	Rec.709; Data Levels Video; 3840×2160

HDR10 TV, 4K (USB / streamer)

Setting	Value
Format	**GDC Matroska** or **GDC MP4** (HDR10 in the container, wide playback); GDC QuickTime for the Apple ecosystem
Codec (Type)	GDC H.265 10-bit (Software x265 Main10), 4:2:0
Preset / Level	slow / **5.1**
Tune	none
Rate Control	**CRF 16–18** (or Target Bitrate 45–80 Mbps)
Keyframe Interval	2 seconds

Setting	Value
HDR10 Metadata	**On**; P3-D65; Peak = your monitor; MaxCLL/MaxFALL measured or 0
In Resolve	Output Color Space Rec.2100 ST2084; Data Levels Video
Audio	PCM in the GDC formats; if the TV does not play PCM, try GDC Matroska or use a Resolve format with AAC/AC-3 (the metadata stays in the stream, but not in the container)
Keep in mind	Do not use 4:2:2 for TV. Check on the real device that HDR10 mode is activated.

HLG TV

Setting	Value
Format	GDC MP4 or GDC Matroska
Codec (Type)	GDC H.265 10-bit (Software x265 Main10)
Preset / Level	slow / 5.1
Rate Control	**CRF 16–18**
HDR10 Metadata	**Off**
In Resolve	Output Color Space Rec.2100 HLG; Data Levels Video

Apple TV, iPhone, iPad (local playback)

Setting	Value
Format	**GDC QuickTime** or **GDC MP4** (hvc1 tag, required by the Apple ecosystem)
Codec (Type)	GDC H.265 10-bit (Software x265 Main10) for HDR; GDC H.265 (x265) 8-bit for SDR
Preset / Level	slow / 5.1 (4K) or 4.1 (1080p)
Rate Control	**CRF 18–20**
HDR10 Metadata	On for HDR10; Off for SDR/HLG
Keep in mind	Avoid 4:2:2 for playback on consumer devices.

10.5 Broadcast television (delivery to a station)

Deliveries to television stations are governed by each station's **technical specification**. The most common requirements are XDCAM HD422 (MXF), ProRes 422 HQ, DNxHD/DNxHR or AVC-Intra: native codecs in Resolve, not in this plugin. **Always follow the station's specification.**

The plugin fits when the station or platform explicitly asks for **H.264 or H.265**, for example for OTT deliveries, viewing archive copies or specifications of the "H.264 High 4:2:2 10-bit" kind. Loudness (for example EBU R128 or ATSC A/85) is set in Fairlight; the plugin does not modify it.

H.264 4:2:2 10-bit delivery for post-production / OTT

Setting	Value
Format	**GDC QuickTime** (or GDC Matroska)
Codec (Type)	GDC H.264 4:2:2 10-bit (Software x264 High 4:2:2)
Preset	slow

Setting	Value
Profile / Level	High 4:2:2 (automatic) / 4.1 (1080p 25–30) or 4.2 (1080p 50–60)
Tune	none
Rate Control	**CRF 12–16** (or Target Bitrate 50–100 Mbps)
Keyframe Interval	1 second
HDR10 Metadata	Off (SDR Rec.709)
In Resolve	Output Color Space Rec.709; Data Levels Video; the resolution and frame rate required by the specification
Audio	24-bit 48 kHz PCM
Keep in mind	Check the exact profile in the specification (many broadcast specifications require an intra-only profile or fixed parameters, which CRF does not guarantee).

10.6 Cinema, festivals, projection

For theatrical projection, **DCP** (JPEG 2000, XYZ space) and **ProRes 4444 XQ**, **DPX** or **EXR** masters are used, all native in Resolve. The plugin is **not** the tool for the cinema master. You use it for viewing files and online delivery.

Festival screener / online submission (1080p)

Setting	Value
Format	MP4 (Resolve) with AAC or QuickTime (Resolve)
Codec (Type)	GDC H.264 (Software x264), 8-bit
Preset / Profile / Level	slow / High / 4.1
Rate Control	**CRF 16–18** (or Target Bitrate 15–25 Mbps)
Keyframe Interval	2 seconds
In Resolve	Rec.709; Data Levels Video; 1920×1080; 24 fps (the film's rate)
Keep in mind	Many festivals ask for 1080p H.264; read the rules before exporting.

HDR review for a client or colorist (with HDR10 in the file)

Setting	Value
Format	**GDC QuickTime** or **GDC MP4**
Codec (Type)	GDC H.265 10-bit (Software x265 Main10)
Preset / Level	medium / 5.1
Rate Control	**CRF 18–20**
HDR10 Metadata	On; the real mastering monitor; MaxCLL/MaxFALL measured or 0
In Resolve	Rec.2100 ST2084; Data Levels Video

10.7 Light master / quality archive

A 10-bit H.264/H.265 master is **lossy** even at a very low CRF. For a lossless master use Resolve's lossless or intermediate codecs (ProRes, DNxHR, FFV1). The recipes below are for **light** masters, useful when space matters.

Light 4:2:2 10-bit master (SDR)

Setting	Value
Format	GDC QuickTime
Codec (Type)	GDC H.264 4:2:2 10-bit (x264) or GDC H.265 4:2:2 10-bit (x265)
Preset	slower
Rate Control	H.264: **CRF 10–14**. H.265: **CRF 12–16**
Keyframe Interval	1 second
Tune	none (grain if the material has grain)
In Resolve	Rec.709; Data Levels Video; native resolution
Audio	24-bit 48 kHz PCM

Light HDR10 4:2:2 10-bit master

Setting	Value
Format	GDC QuickTime or GDC Matroska
Codec (Type)	GDC H.265 4:2:2 10-bit (Software x265 Main 4:2:2 10)
Preset	slower
Rate Control	**CRF 12–16**
Keyframe Interval	1 second
HDR10 Metadata	On; the real monitor; MaxCLL/MaxFALL measured
In Resolve	Rec.2100 ST2084; Data Levels Video
Keep in mind	4:2:2 does not play on ordinary TVs; it is for post-production.

10.8 Review and dailies

Review copy / dailies

Setting	Value
Format	MP4 (Resolve), AAC audio 128 kbps
Codec (Type)	GDC H.264 (Software x264)
Preset	veryfast
Rate Control	**CRF 23–26**
Tune	fastdecode (smooth scrubbing on weak computers)
Keyframe Interval	1 second (fast scrubbing)

Setting	Value
In Resolve	Rec.709; 1280×720 or 1920×1080; timecode included through Data burn-in if needed

10.9 Smallest possible file

Minimum size at acceptable quality (SDR)

Setting	Value
Format	MP4 (Resolve)
Codec (Type)	GDC H.265 (Software x265) 8-bit
Preset	slow
Rate Control	**CRF 27–30**
Keyframe Interval	4–5 seconds
Tune	none
Keep in mind	H.265 saves about a third compared to H.264, but not all devices play it.

10.10 Special kinds of content

Animation, graphics, screen recording

Setting	Value
Codec (Type)	GDC H.264 (x264); for fine colored text: GDC H.264 4:2:2 10-bit
Tune	**animation**
Preset	slow
Rate Control	**CRF 16–18**
Keyframe Interval	2 seconds
Keep in mind	Colored text on a colored background looks cleaner in 4:2:2; but the result looks right only on players that decode 4:2:2.

Film-grain material

Setting	Value
Codec (Type)	GDC H.264 (x264) or GDC H.265 10-bit
Tune	**grain**
Preset	slow
Rate Control	x264: **CRF 18–20**; x265: **CRF 20–22**
Keep in mind	Grain consumes many bits; the file comes out large. You can reduce the grain in Resolve before export if size matters.

Fine gradients and dark scenes (avoiding banding)

Setting	Value
Codec (Type)	A **10-bit** variant (H.265 Main10 or H.264 High10), even for SDR
Preset	slow
Rate Control	CRF 16–18 (x265) or 15–17 (x264)
Advanced Params	`aq-mode=3` (favors dark areas)
Keep in mind	10-bit reduces banding visibly even if the playback screen is 8-bit; but check playback compatibility.

10.11 Fast preview with hardware

VideoToolbox (Mac) or NVENC (Windows)

Setting	Value
Codec (Type)	GDC H.264 (Apple VideoToolbox) / GDC H.265 (Apple VideoToolbox); on Windows the NVENC variants
Rate Control	**Target Bitrate** 8–20 Mbps at 1080p; 25–50 Mbps at 4K
Keyframe Interval	2 seconds
Keep in mind	Preset, Level, Tune and advanced parameters do not apply to hardware. Quality is weaker at the same bitrate than x264/x265: use hardware for review, not for final delivery.

10.12 Frame rate and interlaced material

- Keep the **timeline's frame rate**; do not change it at export. Rate conversions are done in the project settings.
- For web, progressive rates are recommended (24, 25, 30, 50, 60 fps). Interlaced material is converted in Resolve before export when the destination is web.
- The resolution must have an even width (for 4:2:0 also an even height).

11. Advanced parameters (x264 and x265)

The **Advanced Params** field takes options in the form `key=value`, separated by `:`. The plugin passes them straight to x264 or x265, depending on the codec. They are meant for users who know what they change. Always test on a short clip.

> **Warning.** Options written here take priority over those in the panel, including `master-display` and `max-cll`. Do not duplicate the panel's HDR10 settings here. An invalid parameter prevents the export from starting.

11.1 x264 (H.264)

Parameter	What it does	Example	When
aq-mode	Allocates bits by complexity; 3 favors dark areas	`aq-mode=3`	Dark scenes, gradients
aq-strength	Strength of the adaptation	`aq-strength=0.9`	Fine tuning with aq-mode
psy-rd	Preserves the look of detail (rd, trellis)	`psy-rd=1.0,0.15`	Fewer artifacts / more natural detail
ref	Reference frames	`ref=4`	Better quality; watch the level
deblock	Deblocking filter (alpha,beta)	`deblock=-1,-1`	A slightly sharper image
rc-lookahead	Frames analyzed ahead	`rc-lookahead=40`	Better bit distribution
vbv-maxrate + vbv-bufsize	Bitrate cap (kbps)	`vbv-maxrate=15000:vbv-bufsize=30000`	Platforms with a bitrate limit; approximate CBR

11.2 x265 (H.265)

Parameter	What it does	Example	When
aq-mode	As in x264; 3 favors dark areas	`aq-mode=3`	HDR, dark scenes
psy-rd / psy-rdoq	Preserves detail and texture	`psy-rd=2.0:psy-rdoq=1.0`	Finely textured content
rc-lookahead	Frames analyzed ahead	`rc-lookahead=40`	Better bit distribution
bframes	Number of B-frames	`bframes=4`	Better efficiency; slower
no-sao	Disables the SAO filter (less glossy)	`no-sao=1`	A sharper image, with artifact risk
hdr10-opt	Adjusts quantization for HDR10	`hdr10-opt=1`	10-bit HDR10 exports
vbv-maxrate + vbv-bufsize	Bitrate cap (kbps)	`vbv-maxrate=50000:vbv-bufsize=100000`	Platforms with a bitrate limit

11.3 Ready-to-copy examples

CRF with a bitrate cap for social media (H.264):

```
vbv-maxrate=12000:vbv-bufsize=24000
```

Dark scenes and gradients, H.264:

```
aq-mode=3:aq-strength=0.9:rc-lookahead=40
```

10-bit HDR10, H.265, with quantization optimization:

```
hdr10-opt=1:aq-mode=3:psy-rd=2.0:psy-rdoq=1.0
```

Cap for YouTube HDR 4K 30 fps, H.265:

```
vbv-maxrate=60000:vbv-bufsize=120000
```

> **Note.** The number of B-frames is set by the plugin to 2; if you change it here, check the result and compatibility.

12. Quick settings chooser

Destination	Codec (Type)	Format	Key settings
YouTube SDR 1080p	GDC H.264 (x264)	MP4 (Resolve)	slow, High, CRF 18–20, Rec.709
YouTube SDR 4K	GDC H.264 or H.265 (x264/ x265)	MP4 (Resolve)	slow, CRF 18–20 (x264) / 21–23 (x265), Level 5.0–5.2
YouTube HDR10	GDC H.265 10-bit (x265 Main10)	GDC MP4 / QuickTime	slow, CRF 16–18, HDR10 On, Rec.2100 ST2084
YouTube HLG	GDC H.265 10-bit	GDC MP4 / QuickTime	slow, CRF 16–18, HDR10 Off, Rec.2100 HLG
Instagram / TikTok / Facebook	GDC H.264 (x264)	MP4 (Resolve)	slow, High, CRF 20–22, 9:16 or 4:5, Rec.709
Vimeo	GDC H.264 (x264)	MP4 (Resolve)	slow, CRF 17–19
SDR TV 1080p	GDC H.264 (x264)	MP4 (Resolve)	slow, High, Level 4.1, CRF 17–19, AAC
HDR10 TV 4K	GDC H.265 10-bit	GDC Matroska / MP4	slow, Level 5.1, CRF 16–18, HDR10 On
Apple TV / iPhone	GDC H.265 10-bit	GDC QuickTime / MP4	hvc1, CRF 18–20, HDR10 On (for HDR)
Television (station with H.264)	GDC H.264 4:2:2 10-bit	GDC QuickTime	slow, CRF 12–16, Keyframe 1 s, 24-bit PCM
Festival / screener	GDC H.264 (x264)	MP4 (Resolve)	slow, High, Level 4.1, CRF 16–18
Light master	GDC 4:2:2 10-bit (x264 or x265)	GDC QuickTime	slower, CRF 10–16, Keyframe 1 s
Review / dailies	GDC H.264 (x264)	MP4 (Resolve)	veryfast, CRF 23–26, tune fastdecode
Small file	GDC H.265 (x265) 8-bit	MP4 (Resolve)	slow, CRF 27–30, Keyframe 4–5 s
Fast preview	VideoToolbox / NVENC	MP4 (Resolve)	Target Bitrate 8–20 Mbps

13. Troubleshooting

Problem	Probable causes	Fix
I do not see the GDC codecs or formats in Deliver	Plugin not installed; Resolve not restarted; you have the Free version, not Studio; unsupported system	Reinstall the package; close Resolve, wait at least 20 seconds, then reopen it; check for the Studio version
The export does not start	Invalid Advanced Params; odd width or height	Clear Advanced Params; use even dimensions
The image looks washed out or too contrasty	Data Levels on Full for an SDR file; wrong Output Color Space	Data Levels: Video; check Output Color Space
HDR looks washed out on an SDR screen	Normal behavior: the screen does no tone mapping	Check on a real HDR screen; for SDR deliver a separate version
Banding in the sky or gradients	8-bit quantization; CRF too high	Use the 10-bit variant; lower CRF; add `aq-mode=3`; add a little grain in Resolve
File too large	CRF too low; grainy content; keyframes too frequent	Raise CRF; preset slower; use x265; keyframe 2–4 s
File too small or with visible loss	CRF too high; preset too fast	Lower CRF; preset slow; check Target Bitrate
The TV or phone does not play the file	10-bit or 4:2:2 H.264; level too high; PCM audio	Use 8-bit 4:2:0 (or H.265 Main10 4:2:0); Level 4.1; AAC audio through Resolve's format
The platform does not recognize the HDR	Resolve format (no metadata in the container); Output Color Space is not PQ; HDR10 Metadata Off	Use GDC MP4/QuickTime/Matroska; Rec.2100 ST2084; HDR10 Metadata: On
No audio when playing an MP4	PCM audio in MP4 (ipcm), weak support	Use GDC Matroska or GDC QuickTime; or Resolve's MP4 with AAC
The export is very slow	Slow preset; x265; high resolution	Faster preset; hardware for previews; lower the resolution
NVENC variants do not appear on Mac	NVENC exists only with an NVIDIA GPU on Windows	Normal; use VideoToolbox on Mac
"Cannot add video track to clip" message	You chose a GDC format with a codec that is not GDC H.264/H.265	Choose a GDC codec in the GDC format or use a Resolve format

13.1 Checking an exported file

With a free tool such as **MediaInfo** or **ffprobe** you can check what the file contains: codec, profile, bit depth, color tags (primaries, transfer, matrix), HDR metadata (Mastering display, Content light level) and the audio. For a correct HDR10 export you should see the SMPTE ST 2084 transfer, BT.2020 primaries and, in the GDC formats, the mastering and light-level metadata as well.

```
ffprobe -v error -show_entries
stream=codec_name,profile,pix_fmt,color_primaries,color_transfer,color_space:stream_side_data
file.mp4
```

14. Known limits and what has been verified

This is the honest picture of the plugin's state at version 1.7.0. "Verified" means tested through a real export in DaVinci Resolve Studio 21.1 on macOS Apple Silicon, with the resulting file checked.

Feature	Status
x264 and x265, 8-bit 4:2:0	Verified
Apple VideoToolbox H.264 and H.265	Verified
NVIDIA NVENC	Verified in earlier versions on Windows with an NVIDIA GPU; not re-verified at 1.7.0
x264 High10 and x265 Main10 (10-bit 4:2:0)	Verified
x264 and x265 4:2:2 10-bit	Verified (including the order of the color planes)
PQ, HLG, P3-D65, P3-DCI, Rec.709 tags	Verified
HDR10 in the stream (x264 and x265; 4:2:0 and 4:2:2)	Verified on all four combinations
Mastering Primaries: P3-D65	Verified
Mastering Primaries: Rec.2020	**Not tested** in Resolve
GDC QuickTime, GDC MP4, GDC Matroska (x265 4:2:2 10-bit, PQ, HDR10)	Verified, with HDR in the container and PCM audio
H.264 codec in the GDC formats	**Not tested**
"Export Audio" unchecked in the GDC formats	**Not tested**
Windows: loading the plugin	Verified automatically at release
Windows: export in Resolve, including the GDC formats	**Not tested**
Start timecode and markers in the GDC formats	Not implemented
AAC in the GDC formats	Not implemented
Intel Mac, Linux	Not supported

> **Note.** What is marked "Not tested" does not mean it does not work, only that it has not been verified through a real export. If you use it, test on a short clip first.

Annex A. Pre-export checklist

1. The timeline is finished; frame rate and resolution are the intended ones (even width and height).
2. Output Color Space matches the destination (Rec.709, Rec.2100 ST2084 or Rec.2100 HLG).
3. You chose the format (Resolve's or GDC) suited to the audio and metadata you need.
4. You chose the right codec variant: 8-bit 4:2:0 for compatibility, 10-bit for HDR, 4:2:2 only for post-production.
5. Preset: slow for delivery; Level: Auto; Tune: none, unless you have a specific reason.
6. Rate Control: CRF (or Target Bitrate if the platform imposes a bitrate).
7. Data Levels: Video.
8. For HDR10: HDR10 Metadata On, the real mastering monitor, MaxCLL/MaxFALL measured or 0.
9. You exported a short fragment and checked it on the target device (and, if needed, with MediaInfo/ffprobe).
10. Only then do you export the whole film.

Annex B. Glossary

Term	Explanation
CRF	Constant Rate Factor: constant-quality encoding mode; lower number = better quality, larger file
QP	Quantization parameter: how much each frame is compressed
Keyframe / GOP	A keyframe (complete image) / the group of frames between two keyframes
B-frame	A frame coded from the frames before and after it; helps compression
Profile / Level	Set of permitted tools / limits on resolution, frames and bitrate; determines compatibility with devices
Chroma 4:2:0 / 4:2:2	How much color information is kept relative to luminance; 4:2:2 keeps more vertically
Rec.709	SDR standard for HD television; gamma 2.4 (BT.1886)
Rec.2020	The wide color gamut used for HDR and UHD
P3-D65 / P3-DCI	The P3 color gamut with a D65 white point / with the DCI cinema white point
PQ (SMPTE ST 2084)	The HDR transfer function based on absolute luminance; used by HDR10
HLG	Hybrid Log-Gamma: an HDR transfer function compatible with SDR screens; used in television
HDR10	PQ + Rec.2020 + 10-bit + static metadata (mastering display, MaxCLL, MaxFALL)
Mastering display	The characteristics of the monitor you graded on (primaries, luminance)
MaxCLL / MaxFALL	The maximum luminance of any pixel / the maximum average luminance of any frame, over the whole film, in nits
Nit	Unit of luminance (candela per square meter)
SEI	An extra message in the video stream; here it carries the HDR metadata
mdcv / clli	Container boxes that carry the mastering display and the content light level
Container	The file that packages video and audio (MOV, MP4, MKV)
Data Levels (Video/Full)	The value range: Video (16–235 at 8-bit) or Full (0–255)

Annex C. Quick reference for levels

Resolution and rate	H.264 (Level)	H.265 (Level)
1280×720, 30 fps	3.1	3.1
1920×1080, 30 fps	4.0 / 4.1	4.0 / 4.1
1920×1080, 60 fps	4.2	4.1
2560×1440, 30 fps	5.0	5.0
3840×2160, 30 fps	5.1	5.0
3840×2160, 60 fps	5.2	5.1

When you are not sure, leave **Level: Auto**.